Designing for Inclusive Communication:

Insights from a Co-Design Workshop on Live Captioning Systems

Project: Sunva AI
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1. Introduction

Live captioning systems are often designed based on assumptions of fluent, standardised speech and monolingual language use. This framework overlooks the diverse communicative needs of the Deaf, Hard-of-Hearing community, and individuals with speech irregularities. The co-design workshop explored users' lived experiences to identify key challenges, with the insights gathered informing the development of Sunva AI, a live captioning prototype aimed at providing an inclusive and adaptive communication tool.

2. Key Findings from the Co-Design Workshop

- **Validation of Existing Features:** All current Sunva AI features were confirmed to be valuable.
- **Identified Improvements:** The need for enhancements like caption speed control, mixed language caption support, and adaptable TTS tone/style based on context was recognized.
- **Additional Feature Requests:** Suggestions for further features such as multiple speaker identification and accent detection were gathered.
- **Workshop Repetition Unnecessary:** Insights gleaned from the co-design workshops were sufficient, negating the need for further iterations.
- **Impact Assessment:** Sunva AI applies not just to academic and professional contexts, but also enhances communication in everyday situations and tasks.

3. Next steps

1. Engage potential adopters in workplace settings and assess market viability.
2. Existing features in the MVP will be improved based on the co-design workshop participants feedback.
3. A feasibility study with tech experts will validate additional feature requests and identify key features for implementation.
4. Benchmarks will assess the performance of existing capabilities and evaluate privacy, bias, and ethical considerations.
5. Pilots of Sunva AI will be run with selected adopters.

4. Participants

The session included seven participants, aged between 20 to 35 years, representing Deaf and Hard-of-Hearing community and individuals with speech irregularities. Participants represented a range of hearing conditions, including partial to profound hearing loss, and varied levels of English language fluency. Modes of communication included Indian Sign Language (ISL), lip reading, oral speech, and written text. Participants also came from diverse linguistic and professional backgrounds, encompassing both students and professionals, each contributing a broad spectrum of lived experiences relevant to accessible communication.

Participants range of hearing conditions:

1. Deaf persons who do not practice oral communication and prefer sign language as mode of communication.
2. Higher hearing loss, oral communication, lipreading deaf.
3. Deaf, limited lipreading, better communication with a hearing aid, and good communication using sign language.
4. Deaf persons who do not practice oral communication and prefer sign language as mode of communication.
5. Moderate hearing loss, oral communication with clarity, less lip-reading or no lip-reading at all.

5. Methodology

Format: In-person workshop, 3 hours.

The workshop was structured into three sequential co-design activities:

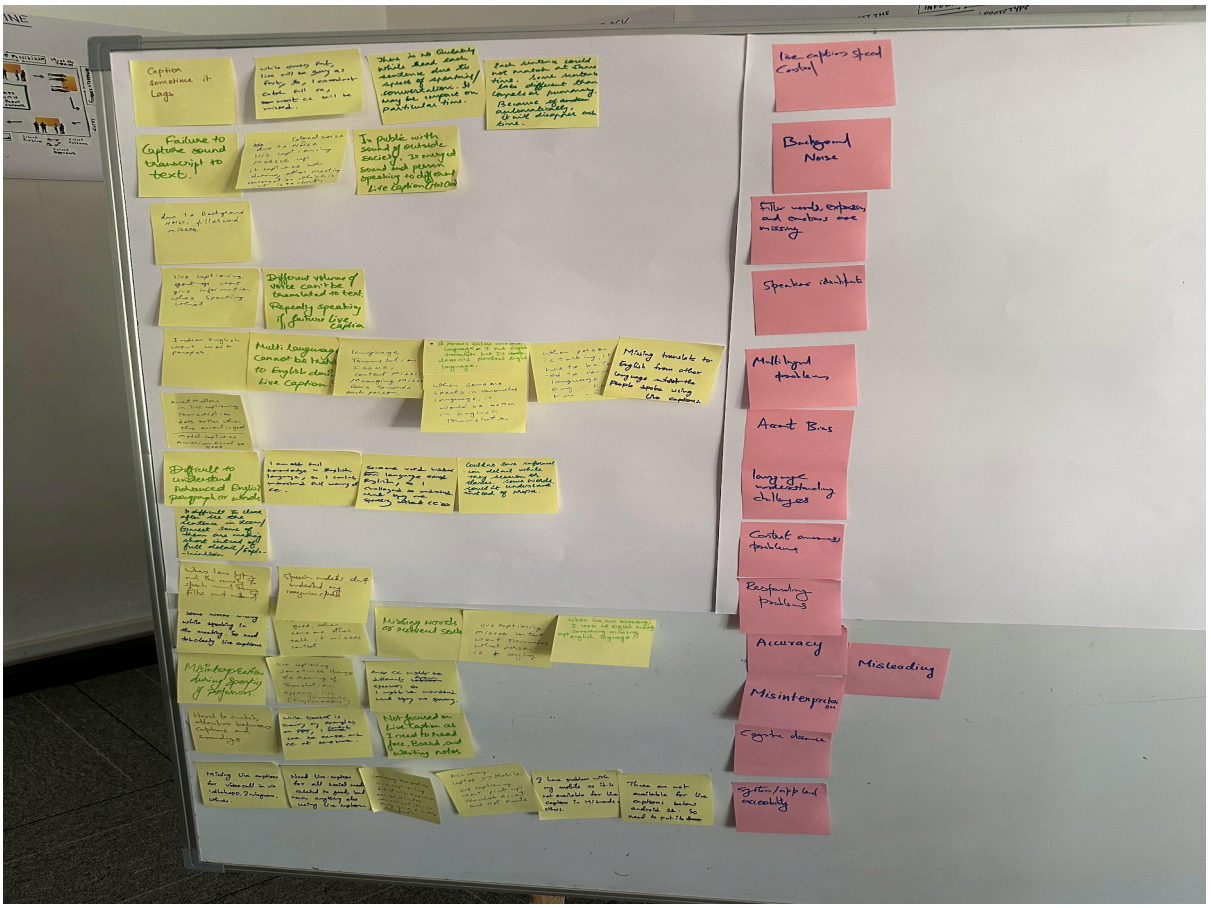
1. Mapping Pain Points in Current Live Captioning Systems
2. Prioritising Feature Enhancements using the MoSCoW Method
3. Evaluating the Sunva AI Prototype using Love, Break, and Fix

Each activity was documented using field notes, discussion, group synthesis activities and participant artefacts.

Activities

Activity 1: Identifying Pain Points in Current Live Captioning Systems

The aim was to surface the nuanced realities and failures of current live captioning technologies as experienced by Deaf, hard-of-hearing, and speech irregularities users.



The raw data, collected via sticky notes, was subjected to thematic analysis.

Thematic analysis results

Themes	Description
Non standard speed of live captions	Users lacked control over the speed and flow of captions. Rapid text made it difficult to keep up or return to missed information.
Background Noise	Irrelevant audio such as filler words and background sound is transcribed, cluttering the captions with non-essential content.
Speaker Identity	The absence of speaker labels led to confusion, especially in multi-person conversations.
Multilingual Problems	Captions failed to handle multilingual utterances or translate across languages, privileging monolingual English speakers.
Accent Bias	Regional, global, or non-native English accents were misinterpreted or corrected inaccurately, reflecting system bias.
Language Understanding Challenges	Users had difficulty following advanced or non-conversational English in captioned form, especially when cognitively overwhelmed.
Context Awareness Problems	Captions lacked situational awareness, failing to reflect speaker emotion, tone, emphasis, or interactional cues.
Responding Problems	Live caption apps help deaf people to listen but responding is still a challenge.
Accuracy vs. Misleading	Captions might be technically “accurate” but still misrepresent meaning or intention. Users found these misleading or emotionally distancing.
Misinterpretation	Systems made semantic or phonetic errors — such as turning speech into the wrong word or misunderstanding context entirely.
Cognitive Dissonance	It is hard to switch attention between captions and the surroundings.
System/app Accessibility	Live captions and text to speech accessibilities are not available in many systems and apps

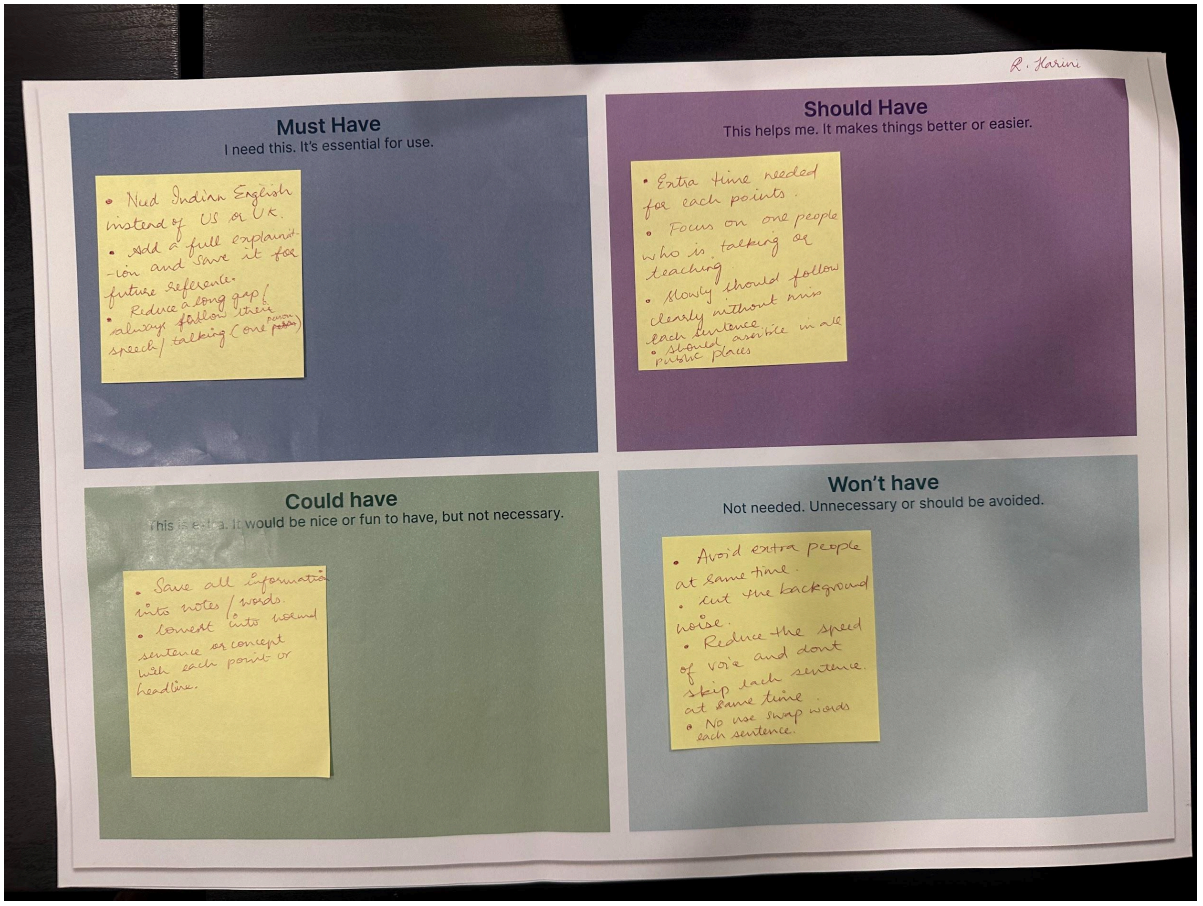
Insights: The thematic analysis of pain points highlighted several key challenges faced by users, as detailed in the table above. **Particularly, multilingual difficulties, language comprehension issues, cognitive dissonance, and challenges in responding emerged as prominent concerns.** Notably, these pain points align with the core issues Sunva aims to address.

Activity 2: Prioritising Features Using the MoSCoW Method

To explore potential features that could enhance the captioning experience, we conducted a participatory activity informed by the challenges identified in Activity 1. Participants were introduced to a set of feature prompts generated using ChatGPT and our AI expertise, aimed at addressing key pain points.

Participants were guided to prioritize these features using the **MoSCoW method.**

- Must Have - Essential for accessibility and usability.
- Should Have - Important but not critical.
- Could Have - Nice to have but not urgent.
- Won't Have - Unnecessary or potentially distracting



Possible features	Must Have	Should have	Could have	Won't have	No of people requested
Text to Speech	✔		✔		3
Background Noise removal	✔				1
Multilingual Support	✔	✔			4
Elaborate Points on demand	✔		✔		3
Create plugins for third party apps	✔				2
Text simplification	✔	✔	✔		7
Multi speaker identification	✔	✔			5
Control live caption speed	✔				7
Mixed language support (Manglish)		✔			
Notification to normal user regarding noise, people interrupting		✔			1
Uncanny voices in TTS		✔	✔		2
Accent detection		✔			2
Third party app support		✔	✔		3
Emotions reference in caption		✔			1

Analysis reveal that the following are the user's priorities:

Prioritized features	Description	No of people requested
Multilingual Support	Live caption applications should support local Indian languages.	4
Text Simplification	Deaf individuals face language understanding problems. The captions should be in simple language for easy understanding and reduced cognitive dissonance.	7
Elaborate points on demand	Click words and phrases in live captions to read the meanings	3
Third party app support	Live caption capabilities should be integrated as a feature in daily used apps like Namma Yatri, Swiggy,...etc.	5
Control live caption speed	By default an ideal live caption app should have near zero latency. But a deaf person should be able to adjust the caption speed based on their reading ability.	7
Text to speech	Text to speech for deaf individuals with non standard speech for responding to normal hearing users.	3
Multi Speaker Identification	Live captions should identify multiple speakers and show their conversations separately.	5
Mixed languages support	Live caption apps should support mixed languages like hinglish, manglish,...etc along with support for multiple languages.	2
Uncanny TTS voices	When some TTS voices are used, normal hearing users have to play it multiple times to understand the content.	3
Accent detection	Live caption apps should detect different accents	2

[The features highlighted in green are already available in Sunva.]

Insights: The MoSCoW activity affirmed that all features currently in Sunva are prioritized by users. The team will assess the remaining features for technical feasibility and ease of implementation, and prioritize them accordingly.

Activity 3: Love – Break – Fix on Sunva AI Prototype

The purpose of this activity is to gather actionable and insightful feedback from participants by exploring their perceptions of value, potential challenges, and suggestions for improving the Sunva AI prototype. This was done through a "Love, Break, Fix" mapping exercise, where participants engaged with the prototype/demo of Sunva AI's live captions, including both text-to-speech and speech-to-text functionalities. From this, we identified key aspects of the prototype that work well and areas that need improvement.

What works	Things to improve
LLM fixing input text before passing to text to speech	UI/UX needs to be improved to help deaf individuals understand how to use the app
Text highlighting	LLM should provide multiple options of possible texts instead of automatically fixing text inputs that goes to TTS
AI automatically switches between simplification and highlighting	Tone and style of TTS output should change based on professional settings or casual settings
Text to Speech	Real time text correction

Insights: The third activity allowed us to pinpoint the successful elements of the Sunva AI prototype and areas for improvement. Participants found all the features in the current version of Sunva AI to be helpful.